Foundation-Guided Active Learning for Efficient Ultrasound Image Segmentation

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ABSTRACT

Medical image segmentation plays a vital role in computer-aided diagnosis and treatment planning. Despite the success of deep learning models, particularly U-Net-based architectures, their performance heavily depends on the availability of large, well-annotated datasets, which are often scarce due to the high cost and expertise required for annotation. Active Learning (AL) presents a promising approach to alleviate the annotation burden by selecting the most informative samples for labeling. However, traditional AL approaches suffer from cold start issues in the early rounds due to the insufficient amount of labeled data. In this work, we propose a foundation-guided deep active learning framework for ultrasound image segmentation. By leveraging pre-trained foundation models, such as DINOv2, we extract rich feature representations that improve both sample selection and model performance, especially during the initial AL rounds. Our approach integrates these models into a U-Net-based segmentation pipeline, demonstrating enhanced data efficiency and segmentation accuracy. By combining the generalization capability of foundation models with task-specific training, our framework provides a practical solution for scalable, label-efficient ultrasound image segmentation. Our code is available at https://github.com/trnKhanh/medical-image-analysis

Keywords: Medical Image Segmentation, Deep Active Learning, Foundation Models, Ultrasound Images

1. INTRODUCTION

Deep learning has become a transformative tool in medical image analysis, delivering state-of-the-art performance in tasks such as classification, detection, and segmentation.[1] However, these models typically require large volumes of high-quality annotated data for training. This poses a significant limitation in medical imaging, where data annotation is not only costly and time-consuming but also demands substantial domain expertise.

Active Learning (AL) offers a promising solution to mitigate the annotation bottleneck by iteratively selecting the most informative samples from a pool of unlabeled data.[2] Despite growing interest in this area, many existing AL strategies struggle during the early rounds, when the labeled set is too small to guide meaningful sample selection, known as the cold-start problem.

In parallel, foundation models trained on large-scale, general-purpose, or domain-specific datasets have demonstrated remarkable generalization capabilities and zero-shot performance across a variety of tasks. Models such as DINOv2[3] and the Segment Anything Model (SAM)[4] offer strong prior knowledge that can potentially enhance sample selection and reduce annotation overhead when integrated into an active learning pipeline.

In this work, we propose a framework for medical image segmentation guided by a foundation of active learning, which is illustrated in Figure 1. We leverage foundation models to accelerate the convergence of our segmentation network within the active learning process. This approach not only improves sample efficiency but also enables the model to reach competitive performance with fewer iterations and reduced manual effort.

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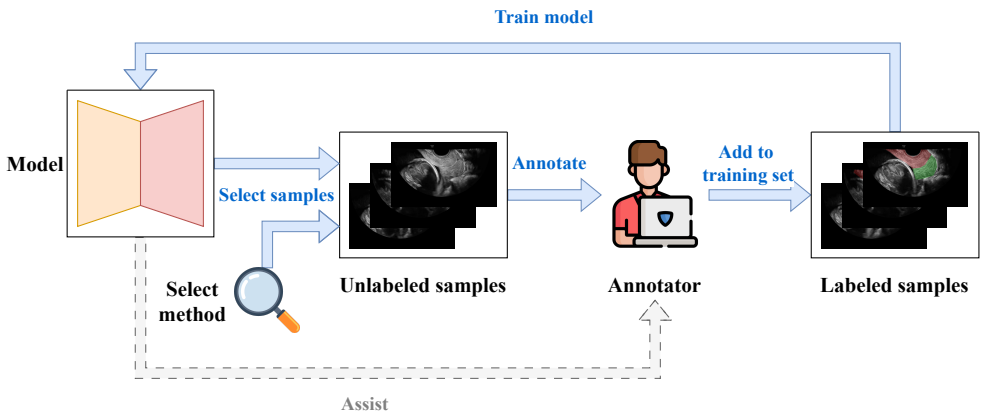


Figure 1: Active learning pipeline

Our contributions are summarized as follows: (1) We propose an active learning pipeline for ultrasound image segmentation under data-scarce conditions; (2) We introduce an algorithm to fuse foundation model embeddings with task-specific (specialist) embeddings to guide the sample selection process; (3) We evaluate the proposed method and compare its performance with existing approaches.

The remainder of this paper is structured as follows: Section 2 provides a brief literature review; Section 3 describes our proposed pipeline in detail; Section 4 presents the experimental setup and comparative results; and Section 5 concludes the paper and discusses our limitations.

2. RELATED WORK

2.1 Ultrasound Image Segmentation

Traditional segmentation methods in ultrasound images, such as thresholding, region growing, active contour models, and level sets[5]—often lack robustness when applied to images with high variability. The advent of deep learning, particularly convolutional neural networks (CNNs), has significantly advanced performance in this field. U-Net[6] and its variants, including Attention U-Net,[7] U-Net++,[8] and nnU-Net,[9] have become foundational architectures due to their ability to capture both local and global contextual information effectively.

2.2 Deep Active Learning

Many deep active learning (DAL) methods employ uncertainty-based approaches, measuring uncertainty through confidence,[10] margin,[11] and entropy.[12] Another category includes representative-based methods, which aim to identify a set of prototypes that effectively represent the dataset. For example, coreset[13] and Cluster-Margin[14] select informative batches that span the data distribution, while BADGE[15] applies a clustering algorithm based on the gradient vectors.

In the medical domain, where data is extremely scarce and annotation is costly, various DAL methods have also been proposed. Different imaging modalities have been explored, including common modalities such as histopathology[16] and X-ray.[17] Additionally, HAL-IA[18] introduced an interactive annotation pipeline for general medical imaging.

However, most DAL research in computer vision focuses on selecting samples under the guidance of a specialist model, which limits performance in the initial stages when the labeled dataset is too small for the model to generalize effectively.

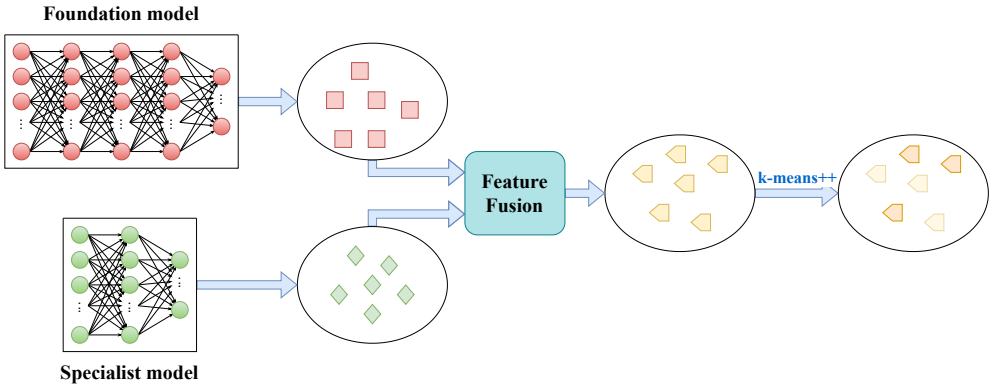


Figure 2: The overview of our active sample selection process. First, embeddings are extracted from both the foundation model and the specialist model. Next, these embeddings are fused using the Feature Fusion module. Finally, the center of each cluster, identified via clustering, is selected as the sample to be annotated.

2.3 Foundation Model

Many foundation models have been developed with promising performance and strong generalization across both general-purpose and medical domains. DINOv2[3] and its variants are trained on massive image datasets, significantly advancing computer vision algorithms and drawing increased attention to foundation models. The Segment Anything Model (SAM)[4] has demonstrated excellent zero-shot performance, greatly enhancing user interaction through prompt-based segmentation. In the medical domain, BiomedCLIP[19] was introduced as a vision-language foundation model.

In this work, we leverage the capabilities of foundation models to improve sample selection in an active learning framework for ultrasound image segmentation. By combining the broad general knowledge of a foundation model with the task-specific expertise of a domain-specialized model, we aim to guide the selection of unlabeled samples more effectively and improve overall model performance with fewer annotations.

3. PROPOSED METHOD

Figure 1 illustrates an overview of our active learning pipeline. The initial set of labeled images is used to train the segmentation model. Subsequently, based on the trained model and the active selection strategy, informative unlabeled samples are selected for annotation (Section 3.2). With assistance from the trained model, experts assign labels to each of the selected samples, which are then added to the training set (Section 3.3). The updated training set is used to retrain the segmentation model. This loop continues until a termination condition is met.

3.1 Preliminary

Given a foundation model $\mathcal{M}_f$, a labeled dataset $\mathcal{D}_{train} = \{\mathbf{x}_i, \mathbf{y}_i\}_{i=1}^{L}$, and an unlabeled dataset $\mathcal{D}_{pool} = \{\mathbf{x}_i\}_{i=1}^{U}$, where $L \ll U$, $\mathbf{x}_i$ is the data vector and $\mathbf{y}_i$ is the onehot-encoded label of the i-th sample, respectively, the process is carried out in T iterations. In the t-th iteration, a batch $\mathcal{Q}^t$ of b samples is chosen from $\mathcal{D}_{pool}^{t-1}$ to be annotated by an oracle and added to $\mathcal{D}_{train}^{t}$, which is then used to retrained the model $\mathcal{M}^t$. The selection of Q^t is based on the specialist model $\mathcal{M}^{t-1}$, the foundation model $\mathcal{M}_f$, and an acquisition function $\alpha(\cdot)$.

3.2 Active Selection

Figure 2 illustrates our active selection procedure. Embeddings from both the foundation model $\mathcal{M}_f$ and the specialist model $\mathcal{M}$ are fused by a feature fusion module. The k-means++ algorithm is then applied to the fused embeddings to identify cluster centers, which correspond to the selected samples.

3.2.1 Foundation Model Feature Fusion

Giving the sample $\mathbf{x}$, we obtain its feature vectors $\mathbf{z}_m$ and $\mathbf{z}_f$ from $\mathcal{M}$ and $\mathcal{M}_f$, respectively. Then, we obtain the fused feature vectors $\mathbf{z}$ by concatenating $\mathbf{z}_m$ and $\mathbf{z}_f$ with a scale factor. Formally, $\mathbf{z}$ is computed as:

$$\mathbf{z} = \mathbf{z}_m \oplus \lambda \mathbf{z}_f, \tag{1}$$

where

$$\lambda = \sqrt{s \cdot \frac{\dim(\mathbf{z}_m)}{\dim(\mathbf{z}_f)}} \tag{2}$$

$\oplus$ is concatenation operation, λ is used to balance the contribution of $\mathbf{z}_m$ and $\mathbf{z}_f$ since the dimension of $\mathbf{z}_f$ could vary tremendously between different foundation models. For DINOv2[3] variations, the output of ViT-B is a 768-dimension vector, while ViT-g outputs a 1536-dimension vector. We introduce a factor s to control the contribution of the foundation representation embeddings.

3.2.2 Sample Selection

The k-mean++ only takes into consideration the distances within the latent space of samples in $\mathcal{D}_{pool}^{t-1}$, hence only optimizes the diversity of sample selection in the context of $\mathcal{D}_{pool}^{t-1}$. As a result, $\mathcal{Q}^t$ might be similar to $\mathcal{D}_{train}^{t-1}$ and the informativeness of chosen samples is reduced significantly. Therefore, we compute the weight vector $\mathbf{w}$ of samples in $\mathcal{D}_{pool}^{t-1}$ to represent the distance between $\mathcal{D}_{pool}^{t-1}$ and $\mathcal{D}_{train}^{t-1}$. The weight $\mathbf{w}_i$ of the sample i-th in $\mathcal{D}_{pool}^{t-1}$ is computed as:

$$\mathbf{w}_i = \min_j ||\mathbf{z}_{pool}^i - \mathbf{z}_{train}^j||, \tag{3}$$

To focus the weight on the distant samples in $\mathcal{D}_{pool}^{t-1}$, we sharpen $\mathbf{w}$ by a sharpening factor β as follows:

$$\hat{\mathbf{w}}_i = \frac{(\mathbf{w}_i)^\beta}{\sum_{j=1}^{U} (\mathbf{w}_j)^\beta} \tag{4}$$

$\hat{\mathbf{w}}$ is then used as the sample weight to initialize the mean of each cluster in the k-means++ algorithm. Therefore, the samples that are most distant from $\mathcal{D}_{train}^{t-1}$ are more likely to be chosen as the clusters' center. Then, $\mathcal{Q}^t$ is the ultimate set of clusters' centers.

3.3 Human-in-the-loop

In the t-th iteration, based on the acquisition function $\alpha(\cdot)$ and the trained model $\mathcal{M}^{t-1}$, we obtain the subset $\mathcal{Q}^t$ from $\mathcal{D}_{pool}^{t-1}$. Then, with the assistance of $\mathcal{M}^{t-1}$, an annotator assigns labels to Q^t by refining the masks generated by $\mathcal{M}^{t-1}$. To reduce the labor intensity of the labeling task, we focus on correcting prominent errors in the predicted masks, such as rough boundaries and mask fragmentation, rather than assigning perfectly precise labels for every sample. This approach decreases both the annotation effort and the level of expertise required to generate medical image labels.

4. EXPERIMENTS

4.1 Datasets

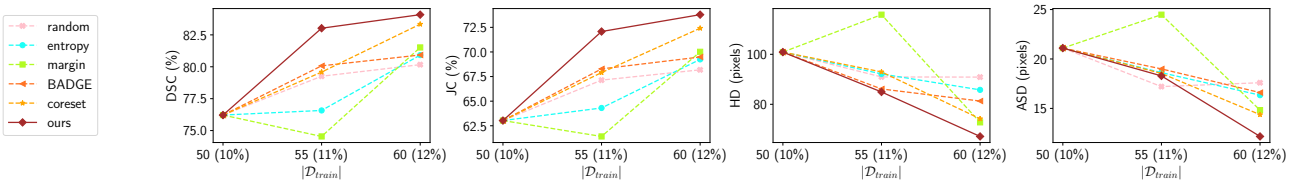
Fetal Ultrasound Grand Challenge 2025

The Fetal Ultrasound Grand Challenge 2025 (FUGC 2025)[20] focuses on cervical segmentation. The top 5 methods on the public leaderboard of the challenge* is demonstrated in Table 1. The FUGC 2025 dataset consists of transvaginal ultrasound images of the cervical anatomical structures. The train, validation, and test sets contain 500, 90, and 300 images, respectively. All images have a resolution of 544×336. The train dataset is divided into a labeled set (50 images) and an unlabeled set (450 images), and all samples in the validation and test sets are labeled.

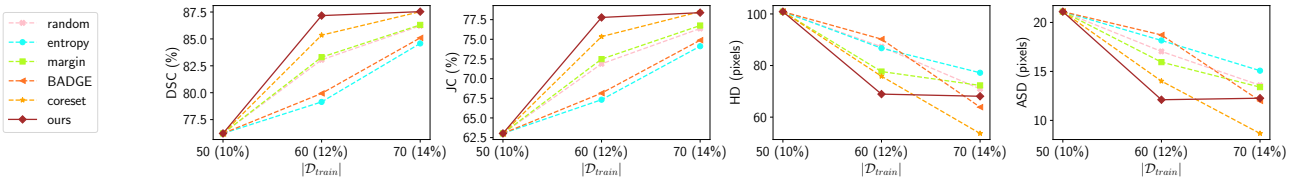
*Public leaderboard is available at https://www.codabench.org/competitions/4781

Table 1: Top-5 teams on the FUGC 2025's main leaderboard (sorted by DSC)

Team	**DSC** (%)↑	**HD** (pixels)↓	**Time** (seconds)↓
zhangxz	90.26	38.89	652.38
tnk2908	90.08	40.39	315.88
AI VIETNAM	87.18	51.02	428.14
stap1e	87.03	47.45	394.44
musarrat	86.29	53.98	509.46
maskoff (Baseline)	71.83	115.41	402.40



(a) Budget $b = 5$



(b) Budget $b = 10$

Figure 3: Quantitative results on the FUGC 2025 dataset

Breast Ultrasound Images Dataset (BUSI)

The Breast Ultrasound Images Dataset (BUSI)[21] comprises a total of 780 ultrasound images, including 437 benign, 210 malignant, and 133 normal cases. Since only the benign and malignant images are annotated, we use a set of 647 labeled images for our experiments. Specifically, 452 images (305 benign and 147 malignant) are used for training, 64 images (43 benign and 21 malignant) for validation, and 131 images (90 benign and 41 malignant) for testing. The average image resolution is approximately 500×500 pixels.

4.2 Experimental Configuration

Our segmentation model $\mathcal{M}$ is based on the U-Net architecture[6] and trained using a composite loss function that combines SoftDice loss and Cross-Entropy loss, similar to nnUnet.[9] We first resize all the images to 256×256 before feeding them to the segmentation model. To train the model, we use an SGD optimizer with a momentum of 0.9, a weight decay of 0.001, and train for a total of 4000 iterations. In the first 200 iterations, we employ the warmup scheme and increase the learning rate linearly to $\text{lr}_0 = 0.01$ to increase the stability of training. Afterward, the learning rate is gradually decreased to 0 using the polynomial function $\text{lr}_i = \text{lr}_0 \cdot (1 - \frac{i}{4000})^{0.9}$. The training batch size is set to 32. For data augmentation, we follow the strategy of nnUNet.[9]

4.3 Evaluation Metrics

To evaluate the performance of the segmentation model, the Dice-Sørensen Coefficient (DSC) and Jaccard Coefficient (JC) are used to measure the region errors, and the Hausdorff Distance (HD) and Average Surface Distance (ASD) are used to measure the boundary errors.

4.4 Quantitative Results

To evaluate the effectiveness of our method, we compared it against several DAL baselines, including random sampling, entropy,[12] margin,[11] BADGE,[15] and coreset.[13] Experiments were conducted on the FUGC 2025 dataset using two budget settings per iteration: $b = 5$ and $b = 10$. To further assess the generalizability of our

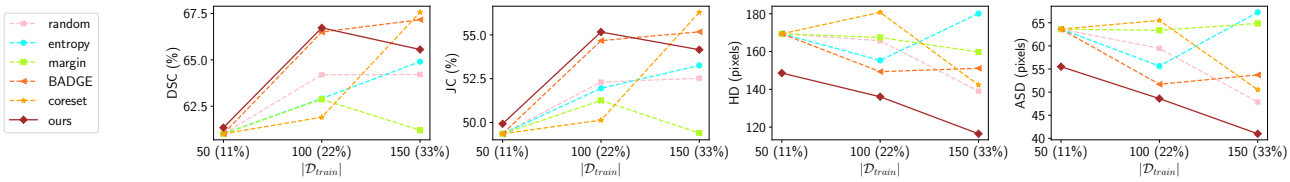


Figure 4: Quantitative results on the BUSI dataset

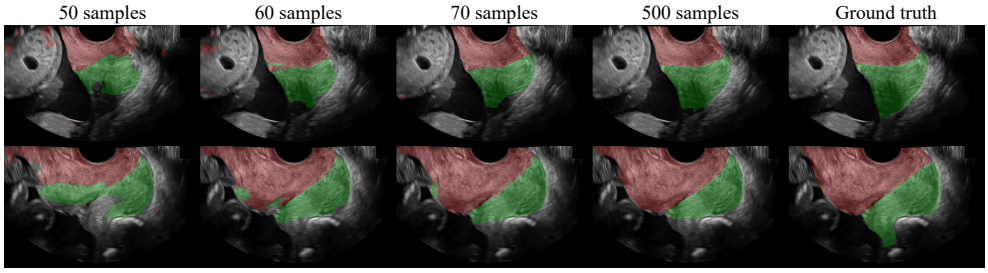


Figure 5: Segmentation masks on the FUGC 2025 dataset across active learning iterations

approach to ultrasound images, we also performed comparisons on the BUSI dataset, using a budget of $b = 50$ per iteration. In all cases, the total number of iterations was set to $T = 2$.

Figure 3 shows the performance of different DAL methods on the FUGC 2025 dataset. Our method outperforms most baselines, particularly for region-based errors and when the training set is small ($|\mathcal{D}_{train}| \leq 60$). Interestingly, a lower budget per iteration tends to result in inferior performance even at the same total training set size. Compared to the top five teams in the FUGC 2025 challenge leaderboard (Table 1), our method, using only 60 labeled samples (12%) out of 500 training samples, achieves performance comparable to the top-three team in terms of DSC, and slightly surpasses them when 10 additional samples are incorporated into the labeled set.

For breast ultrasound images (Figure 4), our method performs comparably to other approaches in terms of region errors during the first two rounds, while significantly outperforming them in terms of boundary errors. However, region performance slightly declines in the final iteration, in contrast to continued improvement in boundary accuracy. A similar trend is observed for the random and margin-based selection methods. We suspect that this decline is due to the inherently noisy boundaries in ultrasound images, which may lead to slight fluctuations in region metrics (approximately 1%) when the active selector and model prioritize boundary refinement.

4.5 Qualitative Results

Figure 5 illustrates segmentation results from models trained on varying numbers of samples. As more samples are added to the training set, the predictions become increasingly accurate, aligning more closely with the ground truth. At $|\mathcal{D}_{train}| = 70$, the model shows significant improvement, demonstrating the effectiveness of the active learning approach. Nevertheless, even the model trained on 500 samples (50 labeled + 450 refined masks) still produces masks with notable errors. This is likely because, during the human-in-the-loop refinement process (Section 3.3), only prominent errors were corrected, without substantially altering the original predictions.

4.6 Ablation Study

Table 2 presents the segmentation model's performance under different settings, measured by iteratively adding 10 and 20 samples to the initial 50 labeled samples from the FUGC 2025 dataset. Incorporating embeddings from the foundation model leads to significant improvement, particularly in the early stages. Additionally, we evaluate the impact of different foundation models on our method's performance by comparing DINOv2[3] (representing the general domain) and BiomedCLIP[19] (representing the medical domain). While DINOv2 achieves slightly

Table 2: Ablation study

$\mathcal{M}_f$	$\mathcal{M}$	s	β	**DSC** (%) ↑		**HD** (pixels) ↓		**JC** (%) ↑		**ASD** (pixels) ↓	
				60	70	60	70	60	70	60	70
✗	✓	-	1.0	80.75	85.50	83.09	78.43	69.19	75.32	17.75	14.92
DINOv2	✓	1.0	1.0	84.87	85.91	71.13	64.73	74.59	76.10	13.60	12.93
BiomedCLIP	✓	1.0	1.0	84.26	87.26	79.29	63.66	73.74	78.05	15.73	**11.18**
BiomedCLIP	✗	-	1.0	81.31	85.39	83.52	74.42	69.91	75.32	18.27	15.76
BiomedCLIP	✓	0.5	1.0	82.83	86.61	79.25	63.29	71.63	77.05	16.96	12.67
BiomedCLIP	✓	2.0	1.0	82.49	86.51	80.48	**62.72**	71.46	76.90	16.89	12.65
BiomedCLIP	✓	1.0	2.0	**87.17**	**87.53**	**68.92**	68.09	**77.78**	**78.39**	**12.11**	12.26
BiomedCLIP	✓	1.0	4.0	80.84	86.22	80.78	68.22	69.51	76.56	16.37	13.07

better results in the initial iteration, BiomedCLIP outperforms it as more samples are added, especially in terms of region-based errors.

Excluding features from the specialist model causes a notable drop in performance, underscoring the importance of combining foundation and specialist embeddings for optimal results. Adjusting the weight of the foundation embeddings also affects overall performance: it declines when the contribution is either too low ($s = 0.5$) or too high ($s = 2$). Lastly, sharpening the sample weight vector in the k-means++ algorithm has a significant positive effect on the region quality of the predicted masks. We find that an intermediate sharpening factor, $\beta = 2$, yields the best performance, while higher values of β can have a negative impact on the results.

5. CONCLUSION

Active Learning plays a crucial role in boosting the performance of deep learning methods, especially in the medical domain where data is scarce and annotation costs are significantly higher than for natural images. To address this, we propose a method that fuses the knowledge of foundation models and specialist models to select unlabeled samples for annotation effectively. Our performance in the FUGC 2025 challenge and on the BUSI dataset demonstrates the efficiency of our approach compared to existing DAL algorithms.

However, our human-in-the-loop scheme currently lacks interactive engagement between humans and the system, which could be greatly enhanced by integrating interactive methods such as SAM-based models. Additionally, reducing and adaptively adjusting the hyperparameters could significantly improve the robustness and usability of our method.

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